

Multimodal interpretable deep learning pipeline for early AD/ADRD screening

1. Abstract

We developed the first multimodal model for early prediction of Alzheimer's disease (AD) and Alzheimer's disease-related dementias (ADRD), integrating voiceprint, timbre, and semantics to improve predictive accuracy and interpretability. Guided by a physician on our team, the model prioritizes clinical applicability and aligns with established diagnostic guidelines. To support practical implementation, we designed a clinical decision support system (CDSS) for deployment in hospitals, health centers, and screening programs. The CDSS enhances transparency in prediction reliability and aids clinicians in making timely, informed diagnostic and treatment decisions. Our model achieved a top-three accuracy ranking on an unseen private dataset, demonstrating its robustness and potential to advance early AD/ADRD diagnosis and intervention.

2. Methodology

Our model architecture, shown in Figure 1, processes the speech signal through three distinct pathways, each with its own prediction model.

(1) Raw Speech Signal Model (Voiceprint Tower): The audio prediction task was formulated as a three-class classification problem using an end-to-end approach. The model includes a built-in preprocessor for fixed f-bank feature extraction and audio volume normalization during pre-training, enabling it to process raw speech signals directly. After evaluating pre-trained models such as Wave2Vec2.0, HuBERT, MetaMMS, and Whisper, we selected the multilingual Whisper medium model for its superior performance, minimal overfitting, and robustness on our limited data. To adapt Whisper's encoder-decoder architecture for classification, we utilized only the encoder, incorporated a linear projection layer to reduce the Transformer's time-step dimensions, averaged the features into a one-dimensional vector, and applied a final linear layer to accomplish the three-class classification.

(2) Text Information Model (Semantic Tower): This pathway focuses on the semantic aspects of speech, leveraging linguistic features to identify cognitive impairments and memory loss commonly associated with Alzheimer's patients. Structured language assessments, such as cookie theft descriptions, verbal fluency tasks, sentence construction, and story recall, are effective in evaluating cognitive status based on response accuracy and coherence. Alzheimer's patients often exhibit response stuttering, forgetfulness, and stammering during these tasks, providing early indicators of the disease. To detect these semantic anomalies, we utilized the state-of-the-art Whisper-Large-V3 speech-to-text model to transcribe speech recordings and fine-tuned a BERT model for classification. This end-to-end approach effectively captures linguistic patterns indicative of cognitive decline, enhancing early detection and screening capabilities.

(3) Frequency information model (Timbre Tower): Numerous studies have identified structured features from specific frequency bands as potential biomarkers for early Alzheimer's detection. Building on this, we transformed raw 1D audio signals into 2D Mel spectrograms via fast Fourier transform to capture frequency

patterns visually. These spectrograms were then processed with the EfficientNet model for three-class classification, enabling the extraction of band-related features. To balance feature richness and prevent overfitting, we followed scaling laws for image models and found EfficientNet-B3 to offer the best trade-off between model size and classification accuracy, effectively identifying subtle timbre differences in Alzheimer’s patients.

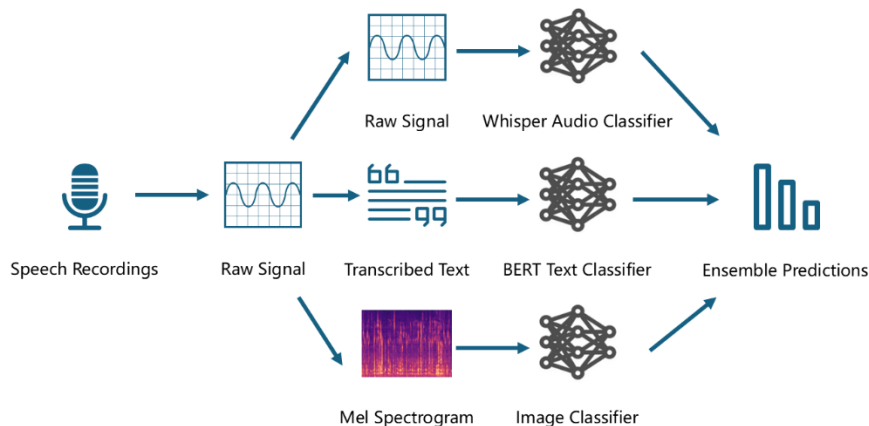


Figure 1. Multimodal framework for AD’s prediction with three processing pathways: audio, text, and spectrogram

Our model computes the final prediction by combining the softmax probabilities from each tower using predefined weights. For the textual information model (semantic tower), we used SHAP to analyze the contribution of each word in the BERT input text. This allowed us to identify key tokens influencing correct predictions and conduct error analysis to pinpoint sources of misjudgment. Word clouds visualizing the most influential words in the cross-validation and test sets provide insights into when the model's predictions are reliable and when they may lack confidence, guiding its clinical application. For the frequency information model (timbre tower), we employed Category Activation Mapping (CAM) to generate heatmaps highlighting areas of the spectrogram that the model focuses on for each prediction. By analyzing spectral activations across validation and test sets, we identified the frequency ranges most relevant to the model's decision-making, offering further guidance for clinical deployment.

To enhance interpretability, we applied various techniques to uncover the model's decision-making process and increase its transparency for clinical use. More detailed methods for applying interpretable information are provided in Section 4: Bias and Ethics Analysis. Additionally, we submitted our comprehensive methodology for model interpretability analysis, including the detailed workflow applied to three representative samples, as part of the Interpretability Track.

3. Performance Analysis

In language-related tasks, 'batch effects' often arise when speakers with similar attributes, such as age, gender, and voice quality, are grouped within the same batch. This can lead to overestimation of model performance on validation sets constructed from the same or mixed batches, while significantly reducing generalization to unseen batches.

To address this, we implemented a group cross-validation strategy using ECAPA-TDNN to extract voiceprint features from the training set. Speakers were clustered into five groups using k-means, which served as cross-

validation folds. Figure 2 shows a 3D scatter plot of these clusters, where distinct colors represent groups. Notably, male and female speakers form prominent sub-clusters, while the clusters exhibit clear spatial separation, reflecting diverse speaker characteristics.

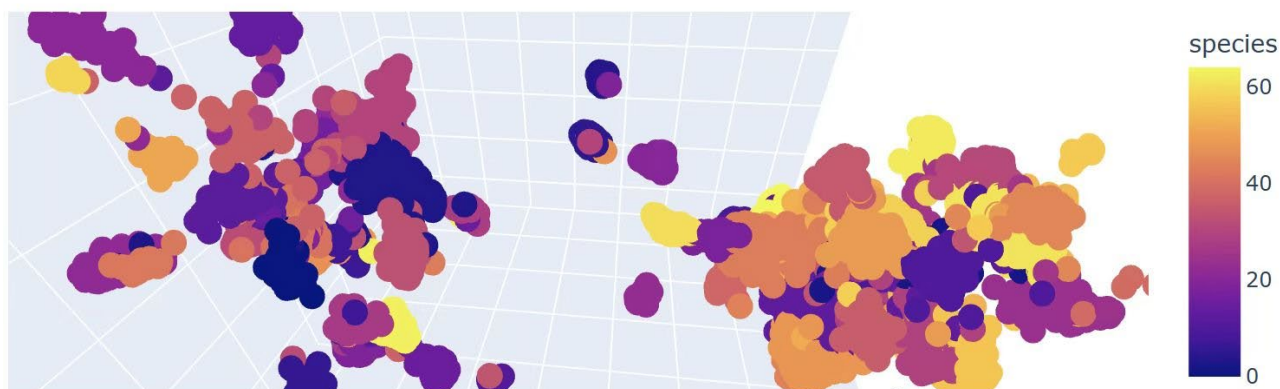


Figure 2. Voiceprint-based clustering of speakers for cross-validation

We used multi-class log-loss as our local evaluation metric, aligning with the competition’s standard. Unlike F1-score or accuracy, log-loss evaluates softmax probabilities directly, capturing the model’s ranking ability without relying on thresholds. Its continuous nature ensures greater stability and representativeness, especially with imbalanced data.

Finally, we performed ablation experiments using 5-fold cross-validation, and the detailed experimental data are shown in the table below:

Methods	Log-loss	delta
Whisper-medium only	0.6687	-
+BERT Ensemble	0.6599	0.0088
+Mel Spec. Ensemble	0.6541	0.0058
+Audio Augmentations	0.6520	0.0021

4. Bias and ethical analysis

We analyzed metadata variables that could bias the model, including 'age,' 'gender,' 'diagnosis,' 'language,' 'education,' and 'corpus.' Due to substantial missing values, 'race' and 'handedness' were excluded. The distributions of the remaining variables are shown in Figure 3.

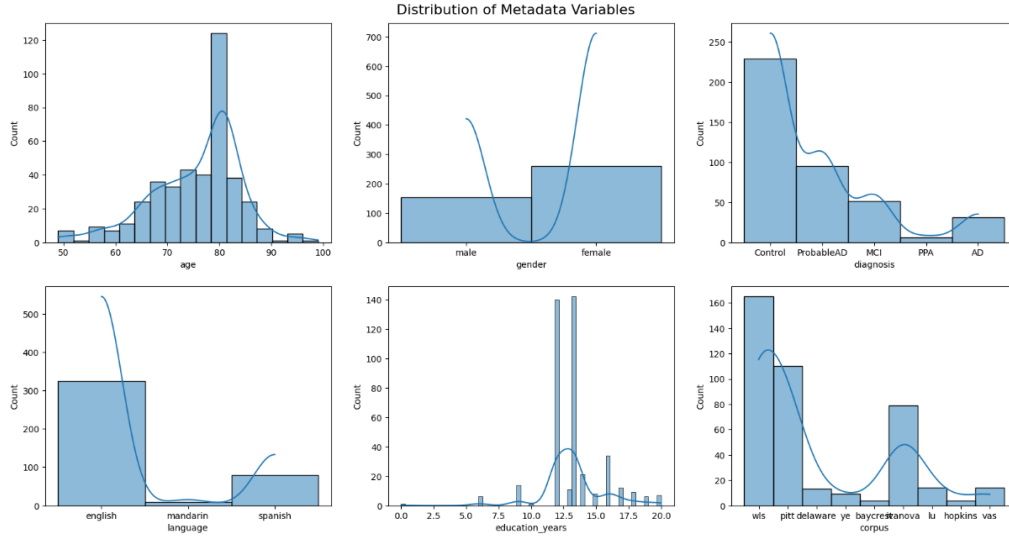


Figure 3. Distribution of metadata variables

To further investigate prediction bias between groups, we analyzed different types of metadata as follows: For the four categorical variables—'gender,' 'diagnosis,' 'corpus,' and 'language'—we calculated grouped log-loss statistics and visualized the differences using violin plots. Additionally, we performed a Kruskal-Wallis H-test for quantitative analysis, with p-values displayed in the titles of the Glocken plots in Figure 4. The results show that p-values for all four variables are less than 0.05, indicating statistically significant differences. These findings highlight notable prediction biases across subgroups, which should be carefully considered in practical applications.

For the two continuous variables, 'age' and 'education,' we conducted a linear regression analysis on the log-loss values, created scatterplots, and calculated Pearson's correlation coefficients. The Pearson r and p-values are displayed in the graph titles. As shown in Figure 4, 'age' exhibits a slight but statistically significant positive correlation with log-loss, while 'education' shows no significant correlation. This suggests that the incidence of ADRD varies significantly with age, potentially increasing prediction difficulty due to greater variance in true values. In contrast, education level does not appear to influence prediction outcomes in this dataset.

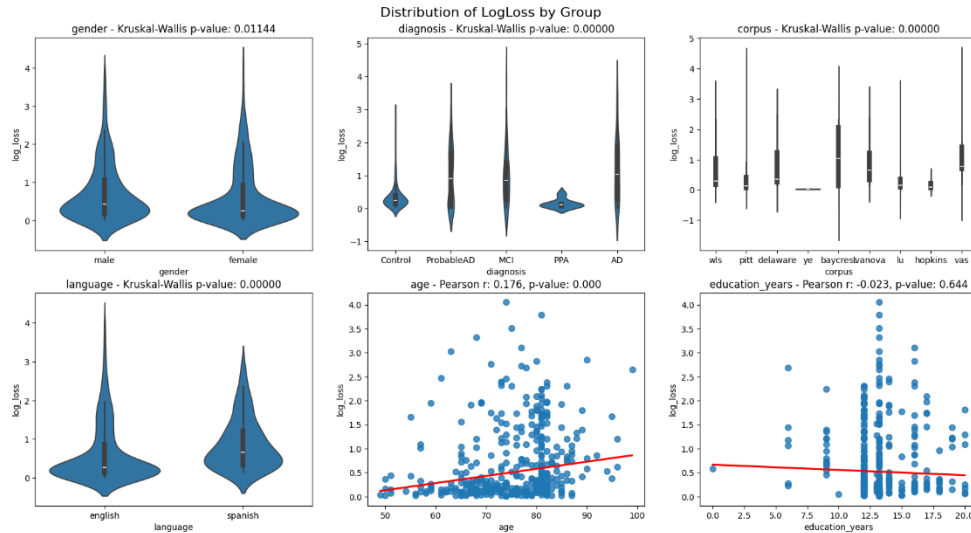


Figure 4. Distribution of log-loss across groups

We transformed the three-class classification problem into two binary screening tasks: Health vs. MCI and Health vs. ADRD, and analyzed ROC curves across subgroups. Given the DeLong test's limitation to paired two-group comparisons, we focused on the binary characteristics of 'gender' and 'language,' each with two subgroups. As illustrated in the graphs A thorough D in Figure 5, for MCI diagnosis, the DeLong test yielded p-values well above 0.05 for English and Spanish, indicating poor pairing and significant differences in ROC performance. In contrast, no significant differences were observed between males and females. For ADRD diagnosis, the gender grouping produced a p-value slightly above 0.05, but overall, no significant differences were detected between gender and language subgroups. These findings highlight notable disparities in model performance across linguistic subgroups for MCI diagnosis, underscoring the importance of subgroup-specific considerations in clinical applications.

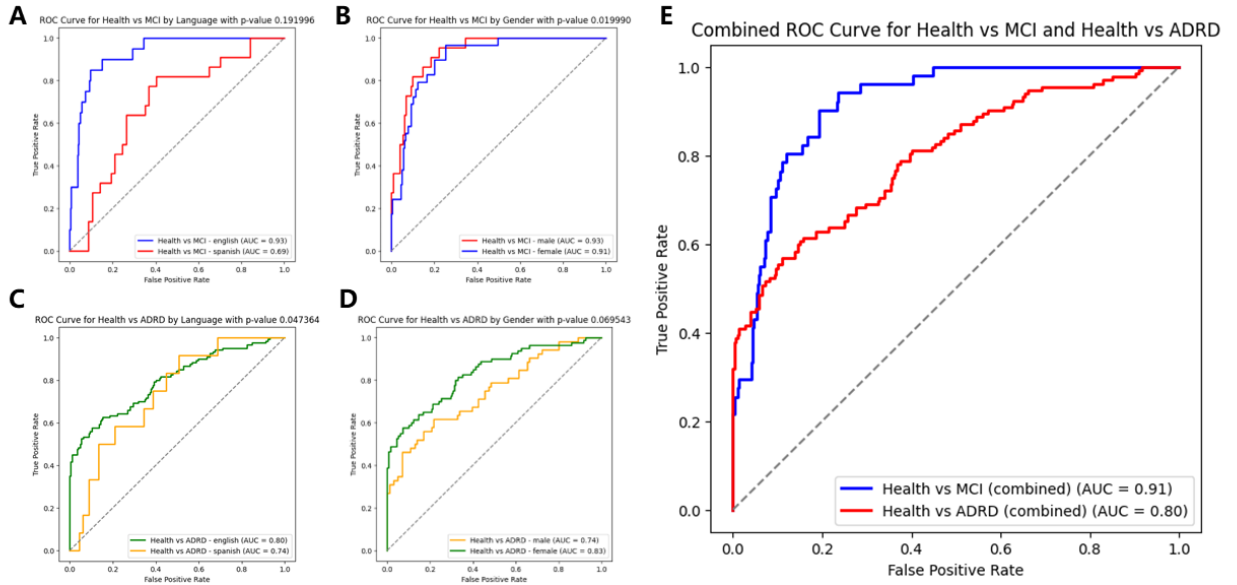


Figure 5. ROC curve analyses for binary screening tasks across subgroups

Analysis of the significant differences between English and Spanish data in the MCI diagnostic task highlights the disparity in pre-training data sizes as the primary cause. Although Whisper and BERT were pre-trained on multilingual datasets, the smaller Spanish corpus limits the model's ability to achieve SOTA performance comparable to English.

To evaluate the model's clinical implications, we conducted ROC analyses, as shown in Figure 5E.

MCI Prediction: A threshold of 0.42 achieves a sensitivity of 1.0, identifying all true MCI patients and making it effective for pre-screening in non-invasive tests like language assessments. A threshold of 0.73 achieves a specificity of 1.0, indicating high confidence in diagnosing MCI for probabilities exceeding this value.

ADRD Prediction: A threshold of 0.20 ensures a sensitivity of 1.0, while 0.88 achieves perfect specificity. Specificity is crucial for ADRD, as follow-up diagnostic tests like biochemical or molecular analyses are often invasive. The model's predictions can guide further testing, reducing unnecessary interventions for low-risk patients and improving clinical efficiency.

To enhance interpretability, we used SHAP to visualize BERT's predictions, providing word-level

explanations for the three-class predictions. Positive contributions are highlighted in pink, and negative contributions in blue. Figure 6 shows a Voice Activation System (VAS) recording predicted as a healthy control (uid: isrt), where BERT relied primarily on text near "call for Alexa." For MCI and ADRD, the model ruled out non-healthy diagnoses based on details like music preferences and the clarity of date interpretation.

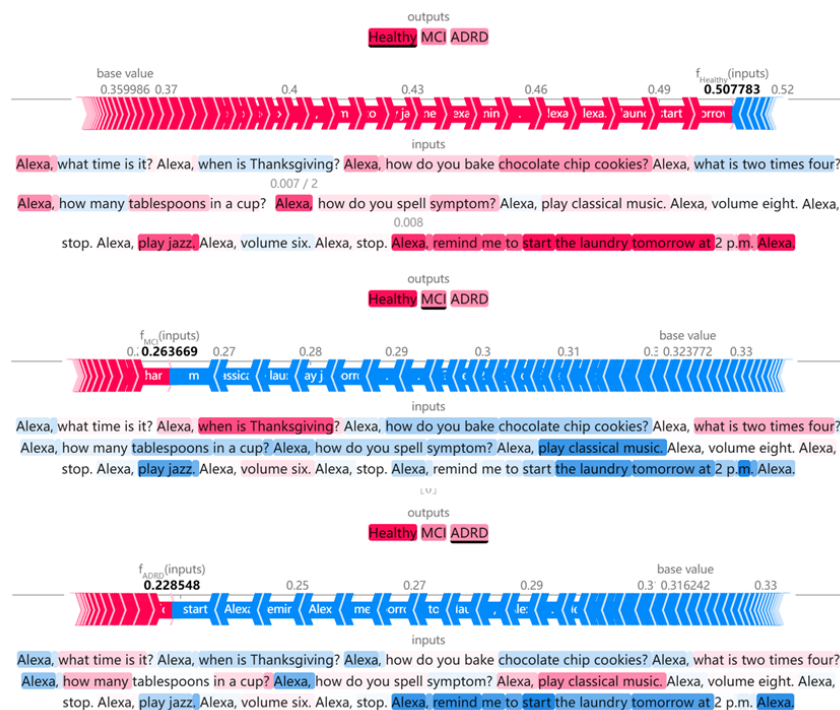


Figure 6. SHAP-based visualization of bert predictions for different diagnostic categories

We analyzed the words contributing most to correct predictions across the three categories and generated word clouds for samples with accurate predictions (Figure 7). Larger words indicate higher frequency, while smaller or absent words suggest lower or no occurrence. SHAP visualizations validate prediction reliability by aligning high-attention words with prominent words in the word cloud. If the focus shifts to smaller or absent words, it may indicate abnormal activation, requiring careful scrutiny of the model's prediction.



Figure 7. Word clouds of key contributors to correct predictions for each diagnostic group

Finally, we applied a commonly used model interpretability technique from computer vision, the Category Activation Map (CAM), to the frequency information model (Timbre Tower). This technique directly highlights the areas of focus within a two-dimensional visual model when generating predictions for a specific category. As shown in Figure 8, the CAM heatmap is overlaid on the spectrogram, revealing the audio frequency and time sequence ranges that the model focuses on. We also analyzed the frequency distribution of the model's focus on the correct category across the entire validation set. Similar to the interpretability of text models, this approach highlights the information ranges on which the model has been well trained. Although data augmentation techniques, such as PitchShift, were used to improve the model's generalization in terms of frequency and timbre, certain characteristic frequency distributions remain and cannot be fully eliminated.

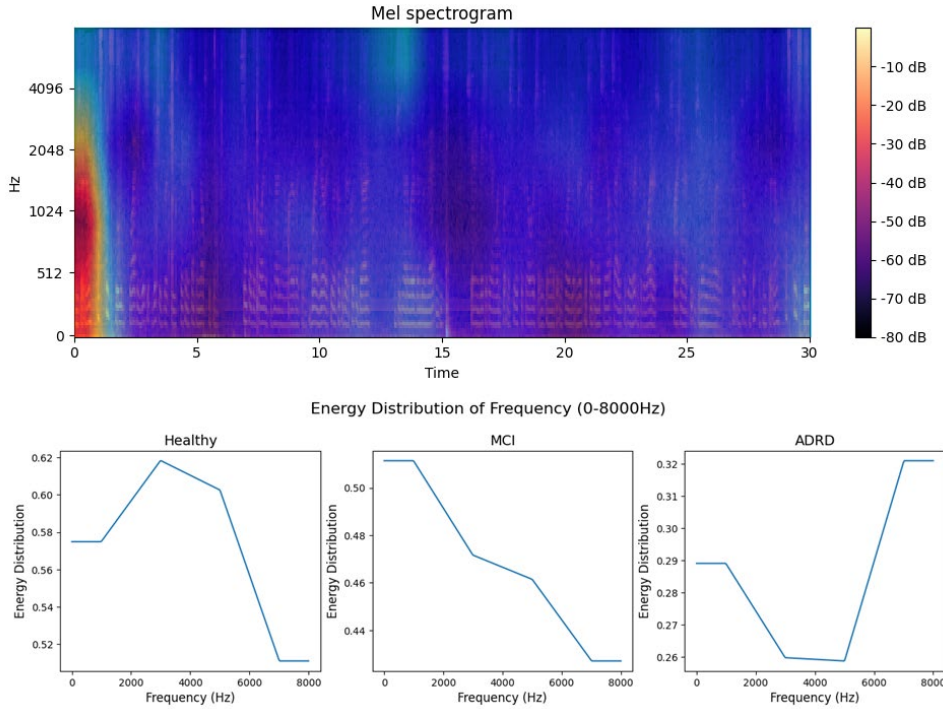


Figure 8. Cam visualization of model focus on spectrograms and frequency distributions across categories

The statistical analysis revealed distinct patterns in the model's focus across frequency ranges for different sample categories. For normal samples, the model predominantly focuses on the mid-frequency range of 2000–6000 Hz. In contrast, the MCI patient model shows greater attention to low-frequency ranges, while the ADRD patient model focuses more on both extremely high and extremely low frequencies. These findings suggest that for unknown samples, predictions are more reliable when the CAM activation on the spectrogram aligns with these statistical characteristics. However, if the activation deviates significantly from these patterns, the model's prediction should be approached with caution or deemed unreliable.

5. Future directions

In the third stage, we plan to access all datasets that were not fully available during the first stage and prototype our model on entirely new datasets, including applications in real clinical environments. The primary objective of this stage is to acquire more diverse and extensive training data. The limited data provided in the second stage posed significant challenges, leading to overfitting and unstable model performance across all competitors. For instance, the Chinese Mandarin dataset in the second stage included only MCI category labels, further

constraining model development. We are confident that our model's multimodal design provides a strong foundation, and its performance can be significantly enhanced with richer datasets. Ultimately, we aim for our model to be implemented in early Alzheimer's screening, demonstrating its effectiveness in real-world clinical settings and contributing to advancements in early detection and intervention.

6. References

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